Keith Geoffrey Ng

Senior Software Engineer | AI and Agentic Engineering | Cloud-Native (AWS, Azure)

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PROFILE

Senior full-stack and cloud-native engineer with nearly 4 years of experience building enterprise systems in finance and insurance. I design serverless, event-driven architectures on AWS and Azure, and I build AI-assisted engineering tooling: multi-agent code review workflows and automated security triage and remediation. Open to senior and lead engineering roles and AI engineering roles.

SELECTED PROJECTS

ai-security-pipeline (github.com/kg-ng/ai-security-pipeline). Node/TypeScript CLI and reusable GitHub Action that runs dependency audits, secret scans, and in-house code-pattern rules, then uses an LLM to triage findings, with a rule-based fallback so it works without an API key. Includes a configurable CI gate that fails builds by severity.

agentic-system (github.com/kg-ng/agentic-system). Reusable multi-agent engineering platform for GitHub Copilot: 32 agent definitions plus skills and path-scoped instructions covering code review, security remediation, architecture governance, DevSecOps, and agent security (OWASP LLM Top 10, MCP hardening). MIT licensed.

ai-recipe (github.com/kg-ng/ai-recipe). Python LLM pipeline that extracts community-tested modifications from recipe reviews and applies them with full citation tracking. Uses batched LLM calls for cost control and a unit test suite with a mocked LLM.

EXPERIENCE

Senior Software Engineer | Australian Finance Group — March 2026 - Present

- Delivered Phase 1 of an automated core banking data pipeline from scratch about 6 months ahead of the planned timeline, accelerated by an in-house multi-agent AI engineering workflow, with few production issues, despite late-surfacing requirements and no access to the CRM's production environment. The pipeline extracts and normalizes the legacy core banking system daily into Amazon Aurora Serverless v2 (PostgreSQL 17), using Drizzle ORM for type-safe schema management, and serves internal applications and third-party platforms.
- Designed a three-layer service architecture (Public, Domain, Supporting) across four AWS accounts with one-way call chains, decoupling public API exposure from core banking data access. Decisions are recorded in ratified Architecture Decision Records.
- Designed event-driven propagation on EventBridge and SQS with per-entity queues, multi-stage dead-letter queues, automated redrive, schema-validated subscribers, and archive/replay for a T+1 sync cycle. Provisioned with AWS CDK (TypeScript) and multi-account pipelines across sandbox, dev, SIT, UAT, and production.
- Architected the multi-agent AI engineering workflow behind that early delivery: 85 agent configurations and 26+ reusable skill pipelines across four service repositories that enforce architectural constraints, TDD, and multi-round automated code review on every pull request.
- Spotted that the company had no automated security scanning and, unprompted, built an AI-driven security pipeline as an interim solution. It scans repositories weekly, triages vulnerabilities by severity and exploitability, and opens verified remediation pull requests, covering direct and transitive dependencies. It has since been adopted for testing across several projects.

Senior Assistant Manager, Senior Full-Stack Engineer | East West Bank Corp. — November 2025 - March 2026

- Led a proof of concept migrating services to Quarkus to improve container startup time and reduce memory footprint.
- Architected a modernization layer that exposes third-party SOAP web services as RESTful APIs, simplifying integration and enabling modern frontend consumption.
- Integrated Trivy into CI/CD for automated container image scanning, and built reusable pipeline boilerplates that standardized deployments and cut setup time for new microservices.
- Resolved 100% of SonarQube-identified vulnerabilities and reached 96% unit test coverage (JaCoCo) on core financial logic.
- Designed the Fund Transfer database schema and logic for high-concurrency transactions, with automated documentation and governance for API contracts.

Senior Software Engineer | Stratpoint Technologies Inc. — November 2024 - October 2025

- Led a team of 5 engineers to deliver a full-cycle procurement system two weeks ahead of schedule, working with business analysts, QA, and stakeholders.

- Built a RESTful Node.js/Express backend with 50+ endpoints for requisition slips, purchase orders, delivery receipts, and payment requests.
- Designed a configurable approval workflow with a custom rule engine supporting dynamic, multi-level, parallel and sequential approvals.
- Implemented role-based access control with 100+ permissions, password resets, temporary passwords, and OTP-based 2FA.
- Built a reusable form component library (React, react-hook-form, zod) that reduced development time by 30%.
- Built data synchronization services, an event-driven email and in-app notification system, and a report service producing multi-page PDF and Excel reports.

Software Engineer | Asurion March 2023 - October 2024

Software Engineer Intern, November 2022 - February 2023

- Built a cost-optimized serverless warranty-tracking application (Go, React Native, AWS Lambda) that reduced EC2 and RDS usage.
- Migrated a legacy application to NestJS (TypeScript), improving maintainability and reducing load times by 20%.
- Built Node.js and React back-office forecasting and analytics applications on a microservices architecture using AWS (EC2, S3, Lambda, RDS).
- Created a Storybook-based React component library, used X-Ray and CloudWatch for tracing and monitoring, and integrated PingFed SSO.
- As an intern, maintained a serverless AWS Lambda application, resolving performance bottlenecks and security vulnerabilities, and built React/Redux features.

SKILLS

Languages and backend	TypeScript, Node.js (NestJS, Express, Fastify), Java (Spring Boot, Quarkus), Go, Python, SQL, MongoDB
Frontend	React, Redux, Next.js, Angular, RxJS, Tailwind CSS, GraphQL, Storybook
AWS	CDK, Lambda, API Gateway, EventBridge, SQS, Aurora, EC2, S3, EKS, CloudFormation, CloudWatch, X-Ray, Cognito, DynamoDB
Azure	Functions, AKS, API Management, Key Vault, Azure SQL, Cosmos DB, Monitor
AI and agentic eng.	multi-agent workflows, custom agents and skills, LLM-assisted triage, AI-driven security remediation, agent security (OWASP LLM Top 10, MCP hardening)
DevOps and security	GitHub Actions, GitLab CI, AWS CodePipeline, Azure DevOps, Docker, Trivy, SonarQube, Grafana, Prometheus, OAuth 2.0, JWT, SSO (Azure AD, Cognito, PingFed)

EDUCATION

B.S. Manufacturing Engineering, Mapua University, 2018 - 2022